Device Info CLI Lab – PBQ Answer Key with Explanations

The lab builds a new network every time it loads or is reset, so the MAC addresses, ports, VLANs and IP addresses change each time. This key shows where every answer comes from, then works one generated problem all the way through.

The network: four PCs are cabled to Switch0. The PCs sit in two VLANs, two PCs in each. Switch0's GigabitEthernet0/1 is the trunk to Router0, which routes between the VLANs on 802.1Q subinterfaces of its GigabitEthernet0/0 (one subinterface per VLAN, numbered with the VLAN ID).

The task: fill in each PC's MAC address, switch interface, VLAN number, VLAN name and IP address. Each PC starts with one of those fields already filled in. The lab answers `help`, `show vlan`, `show interface`, `show mac-address-table`, `show ip arp` and `clear`.

Objectives (owner's "All updated Objectives" doc, Network+ V9): Network implementation – switching technologies (VLANs, interface configuration) and routing technologies (subinterfaces); Network troubleshooting – tools and protocols (command line tools) and switching issues (VLAN assignment).

Where each answer comes from

Field	Device	Command	What to read
Interface	Switch0	`show interface`	The FastEthernet port whose *Port title* line says "Link to PC1" (PC2, PC3, PC4). Type it in the short form the MAC address table uses: `Fa0/2`.
MAC address	Switch0	`show mac-address-table`	The *Mac Address* on that port's row (learned DYNAMIC on the port). Type it in the dotted form shown, such as `0082.3F52.20E3`.
VLAN #	Switch0	`show mac-address-table` or `show vlan`	The *Vlan* column on that same row. `show vlan` confirms it: the port is listed under that VLAN.
VLAN name	Switch0	`show vlan`	The *Name* beside that VLAN number, spelled as shown (for example `IT_Support`, with the underscore).
IP address	Router0	`show ip arp`	The *Address* on the row whose *Hardware Addr* is the PC's MAC. Its *Interface* column is `GigabitEthernet0/0.<VLAN #>`, which confirms the VLAN.

Check	Router0	`show interface`	Each subinterface's *Internet address* is the VLAN's gateway, `192.168.X.1/24`. A PC's IP must be in the same `192.168.X.0/24` network as its VLAN's subinterface.

Solving order

1. **Switch0, `show interface`:** write down which port is the link to each PC.
2. **Switch0, `show mac-address-table`:** for each of those ports, read the MAC address and the VLAN number.
3. **Switch0, `show vlan`:** read each VLAN's name, and check each PC's port is listed under its VLAN.
4. **Router0, `show ip arp`:** find each PC's MAC in the *Hardware Addr* column and read its IP address.
5. **Router0, `show interface`:** check each IP sits in its VLAN subinterface's network.
6. **Check Answers.** A green border is right; a red one is wrong or empty. Fix the red ones from the CLI and check again.

The field each PC starts with is a way in from any direction. Given an IP address, the ARP table gives the MAC, and the MAC table gives the port and VLAN. Given only a VLAN number, two PCs share it, so `show interface` decides which port is which PC.

Traps to watch for

Trap	Why it catches people
Two MAC addresses for one port	`show interface` prints "address is 000a.xxxx.xxxx" for each port. That is the switch port's own hardware address, not the PC's. The PC's MAC is the one the switch learned on that port, in the MAC address table, and the same MAC appears in the router's ARP table.
Long and short port names	`show interface` prints `FastEthernet0/3`; the MAC table and `show vlan` print `Fa0/3`. They are the same port. Type the short form.
Rows are shuffled	The ports in `show interface`, the MAC table and the ARP table are not in PC1 to PC4 order. Match on the values (port, MAC), never on which row something is in.
VLAN 1 and Gig0/1	VLAN 1 (default) holds the unused ports Fa0/5 and Fa0/6 and Gig0/1, the link to Router0. No PC is in VLAN 1.
The router's own network	Router0's GigabitEthernet0/0 has 192.168.1.1/24. That is not either PC VLAN's network; the PCs' networks are on the subinterfaces .<VLAN #>.

Right command, wrong device	The switch keeps the MAC table and the VLANs; the router keeps the ARP table and the IP addresses. Ask either device for the other one's table and it refuses.

> **Pattern to remember:** the switch answers the layer 2 questions (port, MAC address, VLAN); the router answers the layer 3 ones (IP address, gateway). The MAC address is the thread that ties the switch's table to the router's ARP table.

Worked example: one generated problem

Your values will be different every time; the steps are the same. In this problem the lab filled in one field per PC:

PC	Field given	Value given
PC1	MAC address	`00F7.7524.3D58`
PC2	Interface	`Fa0/1`
PC3	IP address	`192.168.21.13`
PC4	VLAN #	`30`

Switch0# show interface

```
FastEthernet0/3 is up, line protocol is up (connected)
  Hardware is FastEthernet, address is 000a.E090.1ED3
  Port title: Link to PC4
FastEthernet0/4 is up, line protocol is up (connected)
  Hardware is FastEthernet, address is 000a.99E9.3156
  Port title: Link to PC3
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is FastEthernet, address is 000a.4238.14D3
  Port title: Link to PC1
FastEthernet0/1 is up, line protocol is up (connected)
  Hardware is FastEthernet, address is 000a.EE26.ADE5
  Port title: Link to PC2
GigabitEthernet0/1 is up, line protocol is up (connected)
  Hardware is GigabitEthernet, address is 000a.f311.2a0d
  Port title: Trunk to Router0
```

Switch0# show mac-address-table

```
Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
  60     00F5.5E2E.7033    DYNAMIC     Fa0/1
  30     0082.3F52.20E3    DYNAMIC     Fa0/4
  60     00F7.7524.3D58    DYNAMIC     Fa0/2
  30     00C7.1861.BD86    DYNAMIC     Fa0/3
```

Switch0# show vlan

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Gig0/1
30   HR                               active    Fa0/4, Fa0/3
60   IT_Support                       active    Fa0/2, Fa0/1
```

Router0# show ip arp

```
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.21.86    26         00C7.1861.BD86  ARPA   GigabitEthernet0/0.30
Internet  192.168.87.42    4          00F7.7524.3D58  ARPA   GigabitEthernet0/0.60
Internet  192.168.87.75    15         00F5.5E2E.7033  ARPA   GigabitEthernet0/0.60
Internet  192.168.21.13    24         0082.3F52.20E3  ARPA   GigabitEthernet0/0.30
```

Router0# show interface

```
GigabitEthernet0/0.30 is up, line protocol is up
  Encapsulation 802.1Q Virtual LAN, Vlan ID 30
  Internet address is 192.168.21.1/24
GigabitEthernet0/0 is up, line protocol is up
  Hardware is CN Gigabit Ethernet, address is 0009.7C10.D101
  Internet address is 192.168.1.1/24
GigabitEthernet0/0.60 is up, line protocol is up
  Encapsulation 802.1Q Virtual LAN, Vlan ID 60
  Internet address is 192.168.87.1/24
```

How each PC was found

PC	Steps
PC1	Given MAC `00F7.7524.3D58`. The MAC table puts it on `Fa0/2` in VLAN `60` (and `show interface` agrees: Fa0/2 is the link to PC1). `show vlan`: VLAN 60 is `IT_Support`. The ARP table maps that MAC to `192.168.87.42` on Gi0/0.60, inside the 192.168.87.0/24 network of subinterface .60.
PC2	Given `Fa0/1`. The MAC table's Fa0/1 row: VLAN `60`, MAC `00F5.5E2E.7033`. VLAN 60 is `IT_Support`. The ARP table maps that MAC to `192.168.87.75`.
PC3	Given IP `192.168.21.13`. The ARP table maps it to MAC `0082.3F52.20E3` on Gi0/0.30. The MAC table puts that MAC on `Fa0/4` in VLAN `30`, which `show vlan` names `HR`; `show interface` agrees that Fa0/4 is the link to PC3.
PC4	Given VLAN `30`, which holds Fa0/4 and Fa0/3. Fa0/4 is PC3's, and `show interface` says Fa0/3 is the link to PC4. The MAC table's Fa0/3 row gives `00C7.1861.BD86`; the ARP table maps it to `192.168.21.86`. VLAN 30 is `HR`.

Answers for this example

PC	MAC Address	Interface	Vlan #	Vlan Name	IP Address
PC1	00F7.7524.3D58	Fa0/2	60	IT_Support	192.168.87.42
PC2	00F5.5E2E.7033	Fa0/1	60	IT_Support	192.168.87.75
PC3	0082.3F52.20E3	Fa0/4	30	HR	192.168.21.13
PC4	00C7.1861.BD86	Fa0/3	30	HR	192.168.21.86

These answers were entered into the lab and Check Answers marked every field right.